

23MAT117

Linear Algebra

Nimmy P

Assistant Professor

Department of Mathematics

Amrita School of Engineering, Bengaluru

Similarity of matrices

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Definition

An $n \times n$ matrix $\hat{A}$ is called similar to an $n \times n$ matrix A if

$$\hat{A} = P^{-1}AP,$$

for some non singular $n \times n$ matrix P .

This transformation, which gives $\hat{A}$ from A , is called a similarity transformation.

Example

$$(1) A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}, P = \begin{bmatrix} 1 & 3 \\ 3 & 6 \end{bmatrix}$$

Ans) The characteristic eqn of A is $|A - \lambda I| = 0$

$$\begin{aligned} \Rightarrow \begin{vmatrix} 1-\lambda & 2 \\ 2 & 4-\lambda \end{vmatrix} &= 0 &\Rightarrow 4 - \lambda - 4\lambda + \lambda^2 - 4 &= 0 \\ &&\Rightarrow \lambda^2 - 5\lambda &= 0 \\ &&\Rightarrow \lambda(\lambda - 5) &= 0 \\ &\Rightarrow \underline{\underline{\lambda = 0, 5}} &&\text{are the eigen values of } A. \end{aligned}$$

$$\text{Now } P^{-1} = \frac{1}{-3} \begin{bmatrix} 6 & -3 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ 1 & 1/3 \end{bmatrix}$$

$$\hat{A} = P^{-1} A P = \begin{bmatrix} -2 & 1 \\ 1 & 1/3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 3 & 6 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 7/3 & 5 \end{bmatrix}$$

Here the eigen values are $0, 5$.

$$\left\{ \begin{bmatrix} -2 & 1 \\ 1 & 1/3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 3 & 6 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 7/3 & 5 \end{bmatrix} \right.$$

$$P^{-1} A P = \hat{A}$$

$\Rightarrow A$ and $\hat{A}$ are similar

A and $\hat{A}$ are having same eigenvalues, which implies similar matrices have equal spectrum.

Properties of similar matrices

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- Trace of $A = 0 + 5 = 0$
- Trace of $\hat{A} = 0 + 5 = 0$
- Determinant of $A = 0 \times 5 = 0$
- Determinant of $\hat{A} = 0 \times 5 = 0$
- Characteristic polynomial of A is $\lambda^2 - 5\lambda$.
- Characteristic polynomial of $\hat{A}$ is $\lambda^2 - 5\lambda$.
- Rank of A is 1.
- Rank of $\hat{A}$ is 1.

Summary

Property	Description
Determinant	A and $P^{-1}AP$ have the same determinant.
Invertibility	A is invertible if and only if $P^{-1}AP$ is invertible.
Rank	A and $P^{-1}AP$ have the same rank.
Nullity	A and $P^{-1}AP$ have the same nullity.
Trace	A and $P^{-1}AP$ have the same trace.
Characteristic polynomial	A and $P^{-1}AP$ have the same characteristic polynomial.
Eigenvalues	A and $P^{-1}AP$ have the same eigenvalues.

- Consider 3×3 matrix A . Let its characteristic equation be

$$|A - \lambda I| = \lambda^3 + a\lambda^2 + b\lambda + c = 0.$$

Then $A^3 + aA^2 + bA + cI_3 = 0_{3 \times 3}$

- Let $\lambda_1, \lambda_2, \lambda_3$ are distinct (assumed) non-zero eigenvalues of matrix A . Then

$$(A - \lambda_1 I)(A - \lambda_2 I)(A - \lambda_3 I) = 0_{3 \times 3}.$$

It can be verified that

- $\text{rank}(A - \lambda_1 I) = 2$
- $\text{rank}(A - \lambda_2 I) = 2$
- $\text{rank}(A - \lambda_3 I) = 2$
- $\text{rank}((A - \lambda_1 I)(A - \lambda_2 I)) = 1$
- $\text{rank}((A - \lambda_1 I)(A - \lambda_2 I)(A - \lambda_3 I)) = 0$

This result can be used to find 2 matrices whose product has a rank less than the rank of both the matrices.

Diagonalization of a matrix

Diagonalization of a Matrix

If an $n \times n$ matrix $\mathbf{A}$ has a basis of eigenvectors, then

$$\mathbf{D} = \mathbf{X}^{-1}\mathbf{A}\mathbf{X}$$

is diagonal, with the eigenvalues of $\mathbf{A}$ as the entries on the main diagonal. Here $\mathbf{X}$ is the matrix with these eigenvectors as column vectors. Also,

$$\mathbf{D}^m = \mathbf{X}^{-1}\mathbf{A}^m\mathbf{X} \quad (m = 2, 3, \dots).$$

Note:

$$\mathbf{D}^2 = \mathbf{D}\mathbf{D} = (\mathbf{X}^{-1}\mathbf{A}\mathbf{X})(\mathbf{X}^{-1}\mathbf{A}\mathbf{X}) = \mathbf{X}^{-1}\mathbf{A}(\mathbf{X}\mathbf{X}^{-1})\mathbf{A}\mathbf{X} = \mathbf{X}^{-1}\mathbf{A}\mathbf{A}\mathbf{X} = \mathbf{X}^{-1}\mathbf{A}^2\mathbf{X}$$

Examples

$$\textcircled{1} \quad A = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$$

$$\text{Here } \lambda_1 + \lambda_2 = 7$$

$$\lambda_1 \lambda_2 = 6$$

$$\Rightarrow \lambda_1 = 6 \text{ and } \lambda_2 = 1$$

So the eigenvalues are 6, 1.

$$\text{for } \lambda = 6, (A - \lambda I)x = 0$$

$$\Rightarrow \begin{bmatrix} -1 & 4 \\ 1 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow -x_1 + 4x_2 = 0$$

$$\Rightarrow x_1 = 4x_2$$

$$\Rightarrow \begin{bmatrix} 4 \\ 1 \end{bmatrix} \text{ is an eigenvector for } \lambda = 6.$$

$$\text{for } \lambda = 1$$

$$\begin{bmatrix} 4 & 4 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow 4x_1 + 4x_2 = 0 \\ \Rightarrow x_2 = -x_1$$

$$\begin{bmatrix} 1 \\ -1 \end{bmatrix} \text{ is an eigenvector corresponding to } \lambda = 1$$

$$\text{Hence, } X = \begin{bmatrix} 4 & 1 \\ 1 & -1 \end{bmatrix} \text{ and } X^{-1} = \begin{bmatrix} 1/5 & 1/5 \\ 1/5 & -4/5 \end{bmatrix}$$

$$D = X^{-1}AX = \begin{bmatrix} 1/5 & 1/5 \\ 1/5 & -4/5 \end{bmatrix} \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 4 & 1 \\ 1 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 6 & 0 \\ 0 & 1 \end{bmatrix}$$

$$2) A = \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix}$$

characteristic equation is $\lambda^2 - 3\lambda + 2 = 0$

Eigen values are 0, 3.

For $\lambda = 0$, $(A - \lambda I)x = 0$

$$\begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow 2x_1 + x_2 = 0$$

$\Rightarrow \begin{bmatrix} 1 \\ -2 \end{bmatrix}$ is an eigenvector for $\lambda = 0$

For $\lambda = 3$, $(A - \lambda I)x = 0$

$$\begin{bmatrix} -1 & 1 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow -x_1 + x_2 = 0$$

$\Rightarrow \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is an eigenvector for $\lambda = 3$

$$X = \begin{bmatrix} 1 & 1 \\ -2 & 1 \end{bmatrix} \quad \text{and} \quad X^{-1} = \begin{bmatrix} 1/3 & -1/3 \\ 2/3 & 1/3 \end{bmatrix}$$

$$D = X^{-1}AX = \begin{bmatrix} 1/3 & -1/3 \\ 2/3 & 1/3 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -2 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 \\ 0 & 3 \end{bmatrix}$$

$$\textcircled{3} \quad A = \begin{bmatrix} -19 & 7 \\ -42 & 6 \end{bmatrix}$$

Characteristic Equation is $\lambda^2 + 3\lambda - 10 = 0$

Eigenvalues are $\lambda = 2, -5$

For $\lambda = 2$, $(A - \lambda I)x = 0$

$$\Rightarrow \begin{bmatrix} -21 & 7 \\ -42 & 14 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow -21x_1 + 7x_2 = 0 \Rightarrow x_2 = 3x_1$$

$\therefore \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ is an eigenvector for $\lambda = 2$.

For $\lambda = -5$, $(A - \lambda I)x = 0$

$$\Rightarrow \begin{bmatrix} -14 & 7 \\ -42 & 21 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \rightarrow x_2 = 2x_1$$

$\therefore \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ is an eigenvector for $\lambda = -5$.

$$\therefore X = \begin{bmatrix} 1 & 1 \\ 3 & 2 \end{bmatrix} \text{ and } X^{-1} = \begin{bmatrix} -2 & 1 \\ 3 & -1 \end{bmatrix}$$

$$D = X^{-1}AX$$

$$= \begin{bmatrix} -2 & 1 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} -19 & 7 \\ -42 & 6 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 3 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 0 \\ 0 & -5 \end{bmatrix}$$

$$④ \begin{bmatrix} -43 & 77 \\ 13 & 93 \end{bmatrix}$$

Eigenvalues are -50 and 100

Eigenvector corresponding to $\lambda = -50$ is $\begin{bmatrix} -11 \\ 1 \end{bmatrix}$

Eigenvector corresponding to $\lambda = 100$ is $\begin{bmatrix} 7 \\ 13 \end{bmatrix}$

$$X = \begin{bmatrix} -11 & 7 \\ 1 & 13 \end{bmatrix} \quad \text{and} \quad X^{-1} = \frac{1}{150} \begin{bmatrix} 13 & 7 \\ 1 & 11 \end{bmatrix}$$

$$\begin{aligned} D &= X^{-1} A X = \frac{1}{150} \begin{bmatrix} 13 & 7 \\ 1 & 11 \end{bmatrix} \begin{bmatrix} -43 & 77 \\ 13 & 93 \end{bmatrix} \begin{bmatrix} -11 & 7 \\ 1 & 13 \end{bmatrix} \\ &= \underline{\underline{\begin{bmatrix} -50 & 0 \\ 0 & 100 \end{bmatrix}}} \end{aligned}$$

$$⑤ A = \begin{bmatrix} 16 & 0 & 0 \\ 48 & -8 & 0 \\ 84 & -24 & 4 \end{bmatrix}$$

Eigenvalues are $\lambda = 16, -8, 4$.

$$\text{for } \lambda = 16, \begin{bmatrix} 0 & 0 & 0 \\ 48 & -24 & 0 \\ 84 & -24 & -12 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{aligned} 2x_1 - x_2 &= 0 \\ 7x_1 - 2x_2 - x_3 &= 0 \end{aligned}$$

If $x_2 = 2$, we get $x_1 = 3$.

$\therefore \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$ is an eigenvector for $\lambda = 16$

$$\text{for } \lambda = -8, \begin{bmatrix} 24 & 0 & 0 \\ 48 & 0 & 0 \\ 84 & -24 & 12 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{aligned} x_1 &= 0 \\ 7x_1 - 2x_2 + x_3 &= 0 \\ x_3 &= 2x_2 \end{aligned}$$

$\therefore \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$ is an eigenvector for $\lambda = -8$.

$$\text{for } \lambda = 4, \begin{bmatrix} 12 & 0 & 0 \\ 48 & -12 & 0 \\ 84 & -24 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$\Rightarrow x_1 = 0$ and $x_2 = 0$
 x_3 are arbitrary

$\therefore \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ is an eigenvector for $\lambda = 4$.

$$\therefore X = \begin{bmatrix} 3 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & 2 & 1 \end{bmatrix} \text{ and } X^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 1 & -2 & 1 \end{bmatrix}$$

$$D = X^{-1} A X$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 1 & -2 & 1 \end{bmatrix} \begin{bmatrix} 16 & 0 & 0 \\ 48 & -8 & 0 \\ 84 & -24 & 4 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 16 & 0 & 0 \\ 0 & -8 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

Theorem

- If an $n \times n$ matrix A has n distinct eigenvalues, then A is diagonalizable.

For example

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 4 & -17 & 8 \end{bmatrix}$$

has three distinct eigenvalues, $\lambda = 4$, $\lambda = 2 + \sqrt{3}$, $\lambda = 2 - \sqrt{3}$. Therefore, A is diagonalizable. Further,

$$P^{-1}AP = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 2 + \sqrt{3} & 0 \\ 0 & 0 & 2 - \sqrt{3} \end{bmatrix}$$

for some invertible matrix P .

$$[-2 \quad 4]$$

$$M = X D X^{-1}$$

Eigenvalues of M are 2, 3

$$\text{Eigenvectors} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$\therefore X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} \text{ and } X^{-1} = \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}$$

$$M = X D X^{-1}$$

$$= \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}$$

$$\therefore M^6 = X D^6 X^{-1}$$

$$= \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 2^6 & 0 \\ 0 & 3^6 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 64 & 0 \\ 0 & 729 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -601 & 665 \\ -1330 & 1394 \end{bmatrix}$$

Spectrum

The set of all eigenvalues of a matrix A is called as the spectrum of A .

Example

$$A = \begin{bmatrix} 100 & 100 \\ 100 & 100 \end{bmatrix}$$

Eigenvalues of A are 0 and 200.

Thus the spectrum of A , $\text{spec}(A) = \{0, 200\}$

Algebraic Multiplicity

- The algebraic multiplicity of an eigenvalue is the number of times that particular eigenvalue occur in the spectrum of A.

Example

$$\text{Let } A = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 3 & 6 & 2 \end{bmatrix}.$$

Eigenvalues of A are 1,1 and 2.

The algebraic multiplicity of 1 is 2

The algebraic multiplicity of 2 is 1

Geometric Multiplicity

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The geometric multiplicity of an eigenvalue λ is the number of linearly independent eigen vectors corresponding to λ .

Example

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Spectrum of $A = \{1, 1\}$

Here $(0, 1)$ and $(1, 0)$ are the linearly independent eigenvectors corresponding to the eigenvalue 1.

Thus the geometric multiplicity of the eigenvalue 1 is 2.

Theorem

THEOREM 5.2.5 Geometric and Algebraic Multiplicity

If A is a square matrix, then:

- (a) For every eigenvalue of A , the geometric multiplicity is less than or equal to the algebraic multiplicity.
- (b) A is diagonalizable if and only if the geometric multiplicity of every eigenvalue is equal to the algebraic multiplicity.

Another Example

Let $A = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$

The characteristic polynomial is $(\lambda - 1)^2$.

$\therefore$ Eigenvalues are $\lambda = 1, 1$.

$\therefore \underline{A.M(1) = 2}$.

Note:

Here Geometric multiplicity is less than Algebraic multiplicity.

Now $(A - \lambda I)x = 0$ for $\lambda = 1$:

$$\Rightarrow \begin{bmatrix} 0 & 0 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow -x_1 = 0$$

$$\Rightarrow \underline{x_1 = 0}$$

$$\therefore x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ x_2 \end{bmatrix} = x_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$\Rightarrow \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ is the eigenvector corresponding to $\lambda = 1$.

$\therefore \underline{G.M(1) = 1}$

Orthogonal Diagonalization

The Orthogonal Diagonalization Problem (Matrix Form). Given an $n \times n$ matrix A , does there exist an orthogonal matrix P such that the matrix $P^{-1}AP = P^TAP$ is diagonal? If there is such a matrix, then A is said to be *orthogonally diagonalizable* and P is said to *orthogonally diagonalize* A .

Conditions for Orthogonal Diagonalizability

If A is an $n \times n$ matrix, then the following are equivalent.

- (a) A is orthogonally diagonalizable.
- (b) A has an orthonormal set of n eigenvectors.
- (c) A is symmetric.

Example

Find an orthogonal matrix P that diagonalizes

$$A = \begin{bmatrix} 4 & 2 & 2 \\ 2 & 4 & 2 \\ 2 & 2 & 4 \end{bmatrix}$$

Solution

The characteristic equation of A is

$$\det(\lambda I - A) = \det \begin{bmatrix} \lambda - 4 & -2 & -2 \\ -2 & \lambda - 4 & -2 \\ -2 & -2 & \lambda - 4 \end{bmatrix} = (\lambda - 2)^2(\lambda - 8) = 0$$

Thus, the distinct eigenvalues of A are $\lambda = 2$ and $\lambda = 8$.

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For $\lambda = 2$

$$(A - \lambda I)X = 0 \Rightarrow \begin{bmatrix} 2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$2x_1 + 2x_2 + 2x_3 = 0$$

Here $r = 1$
 $n = 3$

$$\Rightarrow x_1 + x_2 + x_3 = 0$$

$$\Rightarrow x_1 + b + a = 0$$

$$\Rightarrow \underline{x_1 = -a - b}$$

$$\therefore n - r = 3 - 1 = 2.$$

$$\therefore \text{Let } x_3 = a, x_2 = b.$$

$$\therefore X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -a-b \\ b \\ a \end{bmatrix} = a \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} + b \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

$\therefore \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$ are the eigenvectors corresponding to $\lambda = 2$.

By the previous theorem, A should have orthonormal set of eigenvectors. So by using Gram Schmidt process, $v_1 = u_1 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$

$$v_2 = u_2 - \frac{\langle u_2, v_1 \rangle}{\|v_1\|^2} v_1 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -1/2 \\ 1 \\ -1/2 \end{bmatrix}$$

$$\therefore \underline{q_1 = \frac{v_1}{\|v_1\|} = \begin{bmatrix} -1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{bmatrix}} \quad \text{and} \quad \underline{q_2 = \frac{v_2}{\|v_2\|} = \begin{bmatrix} -1/\sqrt{6} \\ 2/\sqrt{6} \\ -1/\sqrt{6} \end{bmatrix}}$$

For $\lambda = 8$

$$(A - \lambda I)X = 0 \Rightarrow \begin{bmatrix} -4 & 2 & 2 \\ 2 & -4 & 2 \\ 2 & 2 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\sim \begin{bmatrix} -4 & 2 & 2 \\ 0 & -6 & 6 \\ 0 & 6 & -6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$r = 2, n = 3; n - r = 1 \quad \text{Let } x_3 = a$$

$$-4x_1 + 2x_2 + 2x_3 = 0$$

$$-6x_2 + 6x_3 = 0$$

$$-x_2 + x_3 = 0$$

$$-x_2 = -a$$

$$\underline{x_2 = a}$$

$$-4x_1 + 2a + 2a = 0$$

$$-4x_1 + 4a = 0$$

$$\underline{x_1 = a}$$

$$\therefore X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} a \\ a \\ a \end{bmatrix} = a \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$\Rightarrow \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ is the eigenvector corresponding to $\lambda = 8$.

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Finally, using $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_3$ as column vectors, we obtain

$$P = \begin{bmatrix} -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{bmatrix}$$

which orthogonally diagonalizes A . As a check, we leave it for you to confirm that

$$P^T A P = \begin{bmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ -\frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{bmatrix} \begin{bmatrix} 4 & 2 & 2 \\ 2 & 4 & 2 \\ 2 & 2 & 4 \end{bmatrix} \begin{bmatrix} -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 8 \end{bmatrix}$$

Quadratic Forms

Up to now, we have been interested primarily in linear equations; that is, equations of the form $a_1x_1 + a_2x_2 + \dots + a_nx_n = b$. The expression on the left side of this equation, namely, $a_1x_1 + a_2x_2 + \dots + a_nx_n$ is a function of n variables, called a **linear form**.

In a linear form, all variables occur to the first power, and there are no products of variables in the expression.

Here, we will be concerned with **quadratic forms**, which are functions of the form

$$a_1x_1^2 + a_2x_2^2 + \dots + a_nx_n^2 + (\text{all possible terms of the form } ax_ix_j \text{ for } i < j) \quad (1)$$

Quadratic Forms

- For example, a quadratic form in the variables x_1 and x_2 is

$$a_1 x_1^2 + a_2 x_2^2 + a_3 x_1 x_2 \quad (2)$$

and a quadratic form in the variables x_1, x_2 , and x_3 is

$$a_1 x_1^2 + a_2 x_2^2 + a_3 x_3^2 + a_4 x_1 x_2 + a_5 x_1 x_3 + a_6 x_2 x_3 \quad (3)$$

The terms in a quadratic form that involve products of different variables are called the cross-product terms. Thus, in (2) the last term is a cross-product term, and in (3) the last three terms are cross-product terms.

If we follow the convention of omitting brackets on 1×1 matrices, then (2) can be written in matrix

form as $\begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} a_1 & \frac{a_3}{2} \\ \frac{a_3}{2} & a_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ and (3) can be written as $\begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \begin{bmatrix} a_1 & \frac{a_4}{2} & \frac{a_5}{2} \\ \frac{a_4}{2} & a_2 & \frac{a_6}{2} \\ \frac{a_5}{2} & \frac{a_6}{2} & a_3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

Quadratic Form. Symmetric Coefficient Matrix

Let

$$\mathbf{x}^T \mathbf{A} \mathbf{x} = [x_1 \ x_2] \begin{bmatrix} 3 & 4 \\ 6 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 3x_1^2 + 4x_1x_2 + 6x_2x_1 + 2x_2^2 = 3x_1^2 + 10x_1x_2 + 2x_2^2.$$

Here $4 + 6 = 10 = 5 + 5$. From the corresponding *symmetric* matrix $\mathbf{C} = [c_{jk}]$, where $c_{jk} = \frac{1}{2}(a_{jk} + a_{kj})$, thus $c_{11} = 3, c_{12} = c_{21} = 5, c_{22} = 2$, we get the same result; indeed,

$$\mathbf{x}^T \mathbf{C} \mathbf{x} = [x_1 \ x_2] \begin{bmatrix} 3 & 5 \\ 5 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 3x_1^2 + 5x_1x_2 + 5x_2x_1 + 2x_2^2 = 3x_1^2 + 10x_1x_2 + 2x_2^2. \quad \blacksquare$$

Find the symmetric coefficient matrix of the quadratic form given by:

$$(1) 4x_1^2 - 8x_1x_2 + 5x_2^2 \Rightarrow \begin{bmatrix} 4 & -4 \\ -4 & 5 \end{bmatrix}$$

$$(2) -2x_1^2 + 2x_1x_3 + 4x_2x_3 - 9x_3^2 \Rightarrow \begin{bmatrix} -2 & 0 & 1 \\ 0 & 0 & 2 \\ 1 & 2 & -9 \end{bmatrix}$$

$$(3) (x_1 - 2x_2 + 3x_3 - x_4)^2 = x_1^2 + 4x_2^2 + 9x_3^2 + x_4^2 - 4x_1x_2 + 6x_1x_3 - 2x_1x_4 - 2x_2x_3 + 4x_2x_4 - 6x_3x_4$$

$$\Rightarrow \begin{bmatrix} 1 & -2 & 3 & -1 \\ -2 & 4 & -6 & 2 \\ 3 & -6 & 9 & -3 \\ -1 & 2 & -3 & 1 \end{bmatrix}$$

$$(4) (x_1 - x_2)^2 - 4x_3^2 = x_1^2 + x_2^2 - 2x_1x_2 - 4x_3^2 \Rightarrow \begin{bmatrix} 1 & -1 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & -4 \end{bmatrix}$$

$$(5) (x_1 - x_2 + 4x_3)^2 - 4(x_2 - x_4)^2 \Rightarrow \begin{bmatrix} 1 & -1 & 4 & 0 \\ -1 & -3 & -4 & 4 \\ 4 & -4 & 16 & 0 \\ 0 & 4 & 0 & -4 \end{bmatrix}$$

More Examples

$$2x^2 + 6xy - 7y^2 = [x \quad y] \begin{bmatrix} 2 & 3 \\ 3 & -7 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$4x^2 - 5y^2 = [x \quad y] \begin{bmatrix} 4 & 0 \\ 0 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$xy = [x \quad y] \begin{bmatrix} 0 & \frac{1}{2} \\ \frac{1}{2} & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$x_1^2 + 7x_2^2 - 3x_3^2 + 4x_1x_2 - 2x_1x_3 + 6x_2x_3 = [x_1 \quad x_2 \quad x_3] \begin{bmatrix} 1 & 2 & -1 \\ 2 & 7 & 3 \\ -1 & 3 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$



Diagonalizing Quadratic Forms

Diagonalization of Quadratic Forms Let

$$\mathbf{x}^T A \mathbf{x} = [x_1 \ x_2 \ \cdots \ x_n] \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \quad (1)$$

be a quadratic form, where A is a symmetric matrix. We know that there is an orthogonal matrix P that diagonalizes A ; that is,

$$P^T A P = D = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix}$$

where $\lambda_1, \lambda_2, \dots, \lambda_n$ are the eigenvalues of A . If we let

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

where $y_1, y_2, \dots, y_n$ are new variables, and if we make the substitution $\mathbf{x} = P\mathbf{y}$ in (1), then we obtain

$$\mathbf{x}^T A \mathbf{x} = (P\mathbf{y})^T A P \mathbf{y} = \mathbf{y}^T P^T A P \mathbf{y} = \mathbf{y}^T D \mathbf{y}$$

But

$$\begin{aligned} \mathbf{y}^T D \mathbf{y} &= [y_1 \ y_2 \ \cdots \ y_n] \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} \\ &= \lambda_1 y_1^2 + \lambda_2 y_2^2 + \cdots + \lambda_n y_n^2 \end{aligned}$$

which is a quadratic form with no cross-product terms.

Diagonalizing Quadratic Forms

Theorem 9.6.1

Let $\mathbf{x}^T A \mathbf{x}$ be a quadratic form in the variables $x_1, x_2, \dots, x_n$, where A is symmetric. If P orthogonally diagonalizes A , and if the new variables $y_1, y_2, \dots, y_n$ are defined by the equation $\mathbf{x} = P\mathbf{y}$, then substituting this equation in $\mathbf{x}^T A \mathbf{x}$ yields

$$\mathbf{x}^T A \mathbf{x} = \mathbf{y}^T D \mathbf{y} = \lambda_1 y_1^2 + \lambda_2 y_2^2 + \dots + \lambda_n y_n^2$$

where $\lambda_1, \lambda_2, \dots, \lambda_n$ are the eigenvalues of A and

$$D = P^T A P = \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \dots & \lambda_n \end{bmatrix}$$

Conic Sections

We shall now apply our work on quadratic forms to the study of equations of the form

$$ax^2 + 2bxy + cy^2 + dx + ey + f = 0 \quad (2)$$

where $a, b, \dots, f$ are real numbers, and at least one of the numbers a, b, c is not zero. An equation of this type is called a ***quadratic equation*** in x and y , and

$$ax^2 + 2bxy + cy^2$$

is called the ***associated quadratic form***.

Conic Sections: Completing the square and translating

Since the quadratic equation

$$2x^2 + y^2 - 12x - 4y + 18 = 0$$

contains x^2 , x , y^2 , and y terms but no cross-product term, its graph is a conic that is translated out of standard position but not rotated. This conic can be brought into standard position by suitably translating coordinate axes. To do this, first collect x and y terms. This yields

$$(2x^2 - 12x) + (y^2 - 4y) + 18 = 0 \quad \text{or} \quad 2(x^2 - 6x) + (y^2 - 4y) = -18$$

By completing the squares[†] on the two expressions in parentheses, we obtain

$$2(x^2 - 6x + 9) + (y^2 - 4y + 4) = -18 + 18 + 4$$

or

$$2(x - 3)^2 + (y - 2)^2 = 4 \tag{3}$$

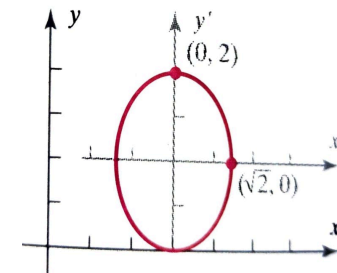
If we translate the coordinate axes by means of the translation equations

$$x' = x - 3, \quad y' = y - 2$$

then (3) becomes

$$2x'^2 + y'^2 = 4 \quad \text{or} \quad \frac{x'^2}{2} + \frac{y'^2}{4} = 1$$

which is the equation of an ellipse in standard position in the $x'y'$ -system. This ellipse is sketched in Figure



Conic Section: Eliminating the cross product term

Eliminating the Cross-Product Term We shall now show how to identify conics that are rotated out of standard position. If we omit the brackets on 1×1 matrices, then (2) can be written in the matrix form

$$\begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} a & b \\ b & c \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} d & e \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + f = 0$$

or

$$\mathbf{x}^T A \mathbf{x} + K \mathbf{x} + f = 0$$

where

$$\mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix}, \quad A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}, \quad K = \begin{bmatrix} d & e \end{bmatrix}$$

Theorem 9.6.2

Principal Axes Theorem for R^2

Let

$$ax^2 + 2bxy + cy^2 + dx + ey + f = 0$$

be the equation of a conic C , and let

$$\mathbf{x}^T A \mathbf{x} = ax^2 + 2bxy + cy^2$$

be the associated quadratic form. Then the coordinate axes can be rotated so that the equation for C in the new $x'y'$ -coordinate system has the form

$$\lambda_1 x'^2 + \lambda_2 y'^2 + d'x' + e'y' + f = 0$$

where λ_1 and λ_2 are the eigenvalues of A . The rotation can be accomplished by the substitution

$$\mathbf{x} = P \mathbf{x}'$$

where P orthogonally diagonalizes A and $\det(P) = 1$.

Example 1: Eliminating the cross-product term

Describe the conic C whose equation is $5x^2 - 4xy + 8y^2 - 36 = 0$.

Solution.

The matrix form of this equation is

$$\mathbf{x}^T A \mathbf{x} - 36 = 0 \quad (7)$$

where

$$A = \begin{bmatrix} 5 & -2 \\ -2 & 8 \end{bmatrix}$$

The characteristic equation of A is

$$\det(\lambda I - A) = \det \begin{bmatrix} \lambda - 5 & 2 \\ 2 & \lambda - 8 \end{bmatrix} = (\lambda - 9)(\lambda - 4) = 0$$

so the eigenvalues of A are $\lambda = 4$ and $\lambda = 9$. We leave it for the reader to show that orthonormal bases for the eigenspaces are

$$\lambda = 4: \quad \mathbf{v}_1 = \begin{bmatrix} 2/\sqrt{5} \\ 1/\sqrt{5} \end{bmatrix}, \quad \lambda = 9: \quad \mathbf{v}_2 = \begin{bmatrix} -1/\sqrt{5} \\ 2/\sqrt{5} \end{bmatrix}$$

Continued in next pa

Example 1

Thus,

$$P = \begin{bmatrix} 2/\sqrt{5} & -1/\sqrt{5} \\ 1/\sqrt{5} & 2/\sqrt{5} \end{bmatrix}$$

orthogonally diagonalizes A . Moreover, $\det(P) = 1$ so that the orthogonal coordinate transformation

$$\mathbf{x} = P\mathbf{x}' \quad (8)$$

is a rotation. Substituting (8) into (7) yields

$$(P\mathbf{x}')^T A (P\mathbf{x}') - 36 = 0 \quad \text{or} \quad (\mathbf{x}')^T (P^T A P) \mathbf{x}' - 36 = 0$$

Since

$$P^T A P = \begin{bmatrix} 4 & 0 \\ 0 & 9 \end{bmatrix}$$

this equation can be written as

$$[x' \ y'] \begin{bmatrix} 4 & 0 \\ 0 & 9 \end{bmatrix} \begin{bmatrix} x' \\ y' \end{bmatrix} - 36 = 0$$

or

$$4x'^2 + 9y'^2 - 36 = 0 \quad \text{or} \quad \frac{x'^2}{9} + \frac{y'^2}{4} = 1$$

which is the equation of the ellipse sketched in Figure 9.6.4. In that figure, the vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ are the column vectors of P . ▲

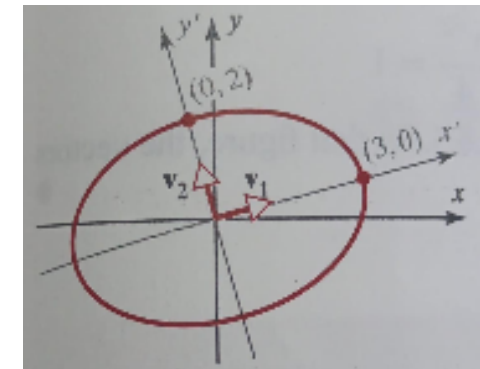


Figure 9.6.4

$$\frac{x'^2}{9} + \frac{y'^2}{4} = 1$$

Example 2

Describe the conic C whose equation is

$$5x^2 - 4xy + 8y^2 + 4\sqrt{5}x - 16\sqrt{5}y + 4 = 0$$

Solution

The matrix form of this equation is

$$\mathbf{x}^T A \mathbf{x} + K \mathbf{x} + 4 = 0 \quad (9)$$

where

$$A = \begin{bmatrix} 5 & -2 \\ -2 & 8 \end{bmatrix} \quad \text{and} \quad K = [4\sqrt{5} - 16\sqrt{5}]$$

Continued in next pa

$$P = \begin{bmatrix} \frac{2}{\sqrt{5}} & -\frac{1}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \end{bmatrix}$$

orthogonally diagonalizes A and has determinant 1. Substituting $\mathbf{x} = P\mathbf{x}'$ into (9) gives

$$(P\mathbf{x}')^T A (P\mathbf{x}') + K(P\mathbf{x}') + 4 = 0$$

or

$$(\mathbf{x}')^T (P^T A P) \mathbf{x}' + (KP) \mathbf{x}' + 4 = 0 \quad (10)$$

Since

$$P^T A P = \begin{bmatrix} 4 & 0 \\ 0 & 9 \end{bmatrix} \quad \text{and} \quad KP = \begin{bmatrix} 20 & -80 \\ \frac{20}{\sqrt{5}} & -\frac{80}{\sqrt{5}} \end{bmatrix} \begin{bmatrix} \frac{2}{\sqrt{5}} & -\frac{1}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \end{bmatrix} = \begin{bmatrix} -8 & -36 \end{bmatrix}$$

(10) can be written as

$$4x'^2 + 9y'^2 - 8x' - 36y' + 4 = 0 \quad (11)$$

To bring the conic into standard position, the $x'y'$ axes must be translated. Proceeding as in Example 5, we rewrite (11) as

$$4(x'^2 - 2x') + 9(y'^2 - 4y') = -4$$

Completing the squares yields

$$4(x'^2 - 2x' + 1) + 9(y'^2 - 4y' + 4) = -4 + 4 + 36$$

or

$$4(x' - 1)^2 + 9(y' - 2)^2 = 36 \quad (12)$$

If we translate the coordinate axes by means of the translation equations

$$x'' = x' - 1, \quad y'' = y' - 2$$

then (12) becomes

$$4x''^2 + 9y''^2 = 36 \quad \text{or} \quad \frac{x''^2}{9} + \frac{y''^2}{4} = 1$$

which is the equation of the ellipse sketched in Figure 9.6.5. In that figure, the vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ are the column vectors of P .

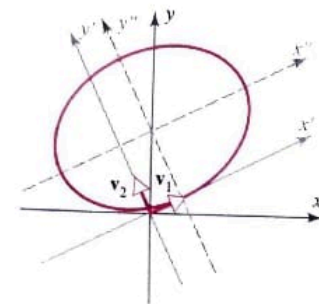


Figure 9.6.5

$$\frac{x''^2}{9} + \frac{y''^2}{4} = 1$$

